Reg. No: | | | | | | | | | | |

SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' GRADE Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2025

B.E – COMPUTER SCIENCE AND ENGINEERING

20CS2E87 – APPLICATION DEVELOPMENT USING MERN STACK

Duration: 2 Hrs 30 min **Maximum: 70 Marks**

PART - A **(5x 1 = 5 Marks)**

1. **CO1** **Which component of the MERN stack is responsible for server-side programming?**
   a) React b) Node.js c) MongoDB d) Redux

2. **CO1** **Which protocol does Express.js primarily use to handle requests?**
   a) HTTP b) FTP c) SMTP d) TCP

3. **CO4** **What does JSX stand for in React?**
   a) Java Syntax Extension b) JavaScript XML
   c) JSON XML d) Java Source eXtension

4. **CO3** **Which Node.js module is used to create a web server?**
   a) fs b) http c) os d) path

5. **CO3** **The state in React can be updated using which method?**
   a) updateState() b) modifyState() c) setState() d) refreshState()

PART - B **(5x 3 = 15 Marks)**

6. **CO1** Illustrate the architecture of the MERN stack and explain the role of each layer in full stack development.
7. **CO2** Apply MongoDB queries to perform CRUD operations on a student collection (insert, update, delete).

8.	**CO3**	Develop a simple Express.js API that connects to a MongoDB collection and fetches all records.	
9.	**CO4**	Explain the use of React components and props to create modular UI elements.	
10.	**CO4**	Apply Redux state management in a React application to manage user session data.	

PART - C **(5 x 10 = 50 Marks)**

Q.No: 11 Compulsory Question

Q.No: 12 to 17 Answer any <u>FOUR</u> Questions

<u>Compulsory Question:</u>

11.	**CO1**		Analyze how the MERN stack architecture supports scalability and maintainability in enterprise web applications.	(10)
12.	**CO1**	i)	Evaluate the performance of MERN applications against traditional multi-tier architectures.	(5)
	CO1	ii)	List the advantages and disadvantages of MERN.	(5)
13.	**CO2**		Apply indexing and aggregation pipelines in MongoDB to optimize query performance for large datasets.	(10)
14.	**CO2**		Analyze how schema design decisions in MongoDB impact data retrieval speed and flexibility.	(10)
15.	**CO3**		Develop a RESTful API using Node.js and Express.js for managing student records with proper error handling.	(10)
16.	**CO3**		Evaluate the benefits of asynchronous programming models for scalable backend design.	(10)
17.	**CO4**		Create a complete MERN stack application for an e-commerce system integrating CRUD, user authentication, and responsive design.	(10)

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